Shaik Mohammad Fardeen

571-386-9025 | shaikfardeen595@gmail.com | LinkedIn | GitHub | Portfolio | Fairfax, VA

AI ENGINEER

EDUCATION

George Mason University, College of Engineering Aug 2025 - May 2027
M.S. Computer Science
Coursework: Intro to Artificial Intelligence, Software Engineering for WWW, Component-Based Software Development, Analysis of Algorithms, Computer Systems and System Programming, Mathematical Foundations of CS, Database Systems, Mobile Immersive Computing, Modern Computer Architecture

Vellore Institute of Technology Sept 2021 - May 2025
B.Tech. Computer Science and Engineering
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, AI & ML

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, R, HTML, CSS
Tools: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter, FastAPI, Azure OpenAI
Core Areas: Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, Statistics, Data Analysis, Feature Engineering

PROFESSIONAL EXPERIENCE

Quadrant Technologies | Cloud AI Intern | Redmond, WA Jun 2026 - Aug 2026
- Developed an explainable candidate ranking system using semantic search and LLMs with adjustable scoring weights across resumes; implemented a blind-review mode showing exactly where each skill matched, ensuring full transparency and trust in AI decisions.
- Built a Screening Assistant with React, FastAPI, and Azure to automate resume parsing; leveraged Azure AI Document Intelligence and Azure OpenAI for intelligent extraction, cutting recruiter manual screening time significantly.

Ethnus Codemithra | Full-Stack Developer Intern Aug 2023 - Nov 2023
- Enhanced UI/UX by refactoring reusable React components and optimizing state management, increasing user engagement 15%; added Jest/React Testing Library coverage and Postman API monitoring, reaching 80% test coverage.

PROJECTS

AI Contract Invoice Intelligence System | Python, TensorFlow, Apache Airflow, Azure Nov 2025
- Built scalable NLP pipelines with Python and TensorFlow to extract and enrich invoice data from unstructured PDFs; processed 150,000+ invoices annually at 98% accuracy, verified through quarterly QA audits.
- Engineered Apache Airflow DAG orchestration for ingestion, text extraction, validation, and storage; automated validation and error detection reduced data-quality issues by 60%.

Personalized Recommendation System | Python, FastAPI, React, Scikit-Learn Oct 2025
- Engineered a multi-model recommendation platform using SVD/matrix factorization, KMeans clustering, and RFM segmentation, with popularity baselining to measure personalization uplift.
- Built a versioned FastAPI backend for recommendations, segmentation, and association rules, plus a React dashboard for model outputs, customer insights, and inventory intelligence.

Loan Default Prediction | PyTorch, XGBoost, SHAP/LIME, Streamlit Feb 2026
- Developed an ensemble of PyTorch neural nets and XGBoost to predict loan defaults, with feature engineering, hyperparameter tuning, and cross-validation.
- Deployed a Streamlit app with SHAP and LIME so banking teams can explain predictions and keep an audit trail for lending decisions.

Autism Detection | Python, TensorFlow, CNN/LSTM, Flask, MySQL Apr 2026
- Designed a CNN/LSTM hybrid for autism spectrum screening and raised baseline accuracy 15% to 91% with learning-rate scheduling and class-weight balancing.
- Built an end-to-end pipeline on 1,500+ labeled medical samples and a Flask + MySQL backend for patient data and model inference.

RESEARCH

Hybrid Algorithm for Parkinson's Prediction | Python, Scikit-Learn, SQL Nov 2025
- Presented a hybrid Random Forest + SVM model at Springer 2024, reaching 88.7% accuracy on the UCI Parkinson's dataset (195 instances) - a 6-8% gain over individual models - by optimizing sensitivity and specificity for early-stage detection.

LEADERSHIP

IEEE Computer Society - Events Head (Mar 2022 - Dec 2024): Led large-scale technical events and hackathons with participation exceeding 700 students.